

March project.py | CMU CS Academy

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1 app.background = gradient(rgb(6, 135, 219), rgb(122, 234, 254), rgb(255, 255, 2
  55), start='bottom')
2 L1 = Label('Match The Dots', 197, 72, size=40, bold=True, italic=True, border=gr
  adient(rgb(69, 119, 234), rgb(56, 171, 228), start='bottom'), fill=None)
3 L2 = Label('You Win!', 197, 72, size=40, bold=True, italic=True, fill=None)
4 L3 = Label('Find the secret extension (hint=orange)', 200, 120, size=20, bold=T
  rue, fill=None, italic=True, border=gradient(rgb(69, 119, 234), rgb(56, 171, 22
  8), start='bottom'))
5 L5 = Label("The average amount of fish species in the Pacific is 3847", 200, 8
  0)
6 L5.visible = False
7 L7 = Label('Wow, thats way too much! This might be dangerous.', 200, 80)
8 L7.visible = False
9 L9 = Label("Wow, looks like your guess is in range, this population is not dang
  erous at all!", 200, 80, size=10)
10 L9.visible = False
11 L10 = Label("WOAH, this population is dangerously low, this could be harmful!",
  200, 80)
12 L10.visible = False
13 linegroup = Group(
14     Line(80, 160, 320, 160),
15     Line(300, 150, 320, 160),
16     Line(300, 170, 320, 160),
17     Line(80, 200, 320, 200),
18     Line(320, 200, 300, 190),
19     Line(320, 200, 300, 210),
20     Line(80, 240, 320, 240),
21     Line(320, 240, 300, 230),
22     Line(320, 240, 300, 250),
23 )
24
25 c1 = Circle(40, 160, 15, fill=rgb(246, 192, 63), border=gradient(rgb(246, 192,
  63), 'gold', 'white', start='right'))
26 c2 = Circle(40, 200, 15, fill=rgb(246, 192, 63), border=gradient(rgb(246, 192,
  63), 'gold', 'white', start='right'))
27 c3 = Circle(40, 240, 15, fill=rgb(246, 192, 63), border=gradient(rgb(246, 192,
  63), 'gold', 'white', start='right'))
28 dots = Group(c1, c2, c3)
29
30
31 cgroup = Group(
32     Circle(360, 160, 15, fill=None, border=gradient(rgb(246, 192, 63), 'gold',
  'white', start='right')),
33     Circle(360, 200, 15, fill=None, border=gradient(rgb(246, 192, 63), 'gold',
  'white', start='right')),
34     Circle(360, 240, 15, fill=None, border=gradient(rgb(246, 192, 63), 'gold',
  'white', start='right')),
35 )
36
37 fishgroup = Group (
38     Polygon(240, 320, 320, 280, 320, 360, fill =rgb(252, 114, 15), border='blac
  k'),
39     Oval(200, 320, 150, 80, fill=rgb(252, 114, 15), border='black'),
40     Line(182, 284, 182, 358),
41     Circle(160, 307, 16, fill='black'),

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42     Circle(164, 304, 8, fill='white'),
43     Circle(161, 335, 8, fill=rgb(252, 114, 15), border='black'),
44     Circle(199, 300, 13, fill=rgb(252, 114, 15), border='black'),
45     Circle(199, 329, 13, fill=rgb(252, 114, 15), border='black'),
46     Circle(230, 318, 13, fill=rgb(252, 114, 15), border='black'),
47     Rect(150, 326, 20, 10, fill=rgb(251, 114, 15)),
48 )
49
50 Button2 = Group(
51     Label('Click for more speech!', 280, 320),
52     Circle(280, 320, 60, fill=None, border='black'),
53 )
54
55 Button2.visible = False
56
57 def checkDots(mouseX, mouseY):
58     if (c1.visible == False and c2.visible == False and c3.visible == False):
59         L2.border = gradient(rgb(69, 119, 234), rgb(56, 171, 228), start='botto
60 m')
61         L1.visible = False
62         if (L3.visible == False):
63             L2.border = None
64         if (c1.hits(mouseX, mouseY) == True):
65             c1.centerY = mouseY
66             c1.centerX = mouseX
67         if (c2.hits(mouseX, mouseY) == True):
68             c2.centerX = mouseX
69             c2.centerY = mouseY
70         if (c3.hits(mouseX, mouseY) == True):
71             c3.centerX = mouseX
72             c3.centerY = mouseY
73         if (cgroup.hits(c1.centerX, c1.centerY) == True):
74             c1.visible = False
75             c1.centerX = 40
76         if (cgroup.hits(c2.centerX, c2.centerY) == True):
77             c2.visible = False
78             c2.centerX = 40
79         if (cgroup.hits(c3.centerX, c3.centerY) == True):
80             c3.visible = False
81             c3.centerX = 40
82         else:
83             pass
84
85 def onMouseDrag(mouseX, mouseY):
86     checkDots(mouseX, mouseY)
87
88
89 def showButton2 ():
90     Button2.visible = True
91     L5.visible = False
92
93 NumberList = []
94
95
96 def onMousePress(mouseX, mouseY):

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97     if (fishgroup.hits(mouseX, mouseY) == True):
98         fishgroup.centerY = 200
99         cgroup.visible = False
100        dots.visible = False
101        linegroup.visible = False
102        L1.visible = False
103        L2.border = None
104        L3.visible = False
105        L5.visible = True
106        Button2.visible = True
107        if (Button2.hits(mouseX, mouseY) == True):
108            guessmessage = app.getTextInput("What do you think the average octopus
species population is (enter guess in number form):")
109            guess_int = int(guessmessage)
110            Button2.visible = False
111            L5.visible = False
112            showButton2 ()
113            if (guessmessage != None and guessmessage != ""):
114                guess_int = int(guessmessage)
115                if (guess_int in NumberList):
116                    print("You already guessed this :(")
117                else:
118                    NumberList.append(guess_int)
119                    if (guess_int > 300):
120                        L7.visible = True
121                        L10.visible = False
122                    elif (guess_int < 300 and guess_int > 0):
123                        L10.visible = True
124                        L7.visible = False
125                    elif (guess_int == 300):
126                        L7.visible = False
127                        L10.visible = False
128                        L9.visible = True
129            else: pass
```